Collaboration gaps in citation networks: a community detection and hypergraph approach

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ABSTRACT

Research institutions often miss internal collaboration opportunities, even among researchers working on related topics. Here, we present a framework to reveal such latent connections within academic institutions by combining community detection and hypergraph analysis on citation networks. Using OpenAIRE data[6], we analyse over 57,000 research products from the University of Pisa since 2020.

We build both a direct citation graph and an expanded graph incorporating bibliographic-coupling edges weighted by Jaccard similarity. Applying Leiden, InfoMap, and ANGEL to both graphs, we identify 18 merge events among communities with $\geq$ 20 nodes. InfoMap independently confirms nearly all of them; ANGEL, which allows overlapping membership, captures boundary structure through fragmentation rather than merging.

Most merges span multiple FOS sub-disciplines—a proportion that depends on the resolution parameter γ, since lower values produce larger communities that mechanically cover more labels. Extending to a hypergraph—where each external reference forms a hyperedge linking all Pisa papers that cite it—we find that co-occurrence in hyperedges correlates strongly with citation: a degree-preserving null model yields $z = 284.5$ ($p < 0.01$).

Extracting gap candidates yields over two million paper pairs sharing references but lacking direct citation. Among the top 50 by debiased score, 72% connect papers in the same broad discipline; among the top 100, 92% fall within the same Leiden community on the expanded graph (random baseline: 0.3%). The two representations—pairwise communities and higher-order co-occurrence—converge on the same structure without being designed to agree. Whether this proximity translates into actionable collaboration requires validation that citation topology cannot provide.

KEYWORDS

Social Network Analysis, Community Detection, Hypergraph, Citation Networks, Research Collaboration, Institutional Networks, OpenAIRE, Collaboration Gaps, University Networks

1 INTRODUCTION

Researchers at the same institution often work on closely related problems without knowing it. We use citation network analysis to surface these hidden overlaps at the University of Pisa, constructing two graphs from publications in the OpenAIRE Research Graph [6]: G_{Cit}, encoding direct citations among 55,078 UniPi papers, and G_{BC}, which adds bibliographic-coupling edges—Jaccard-weighted links between papers sharing external references—reaching 57,603 nodes after isolate removal brings in papers connected only through shared references.

Our central question is whether comparing community structure across G_{Cit} and G_{BC}—and independently through a hypergraph—can surface collaboration opportunities that pairwise citation misses. We apply three community detection algorithms: Leiden and InfoMap (crisp partitions) and ANGEL (overlapping membership). Field of Study labels [5] serve both as independent validation and as the basis for classifying merge events as intra- or cross-disciplinary; they cover 76.9% of nodes, and the unlabelled 23.1% is a limitation affecting all disciplinary classifications here.

Pairwise edges cannot represent multi-way relationships. We therefore build a hypergraph in which a single hyperedge connects all UniPi papers citing the same external reference. The two analyses yield a split picture: most Leiden merge events span multiple FOS sub-disciplines, while the top hypergraph gap candidates are predominantly intra-disciplinary. A degree-preserving null model confirms that hyperedge co-occurrence predicts citation far beyond what degree structure alone can explain ($z = 284.5$, $p < 0.01$).

This report is organized as follows:

- Section 2 — data collection and graph construction;
- Section 3 — topology characterization of graphs;
- Section 4 — community detection on both graphs;
- Section 5 — hypergraph analysis;
- Section 6 — cross-method synthesis;
- Section 7 — limitations and possible extensions.

2 DATA COLLECTION AND GRAPH CREATION

The institutional scope was fixed by querying the OpenAIRE organizations endpoint [1, 6] for entities matching `Pisa`. Of the 50 candidates returned, 13 were confirmed as relevant through manual inspection: the University of Pisa,

constituent departments, specialized research centers, and the Azienda Ospedaliero Universitaria Pisana. All subsequent retrieval relies on their unique organizational identifiers, sidestepping the ambiguities of string-based affiliation matching.

Retrieval and preprocessing

Filtering the OpenAIRE Graph API by the 13 organizational identifiers yielded 264,930 research products. Records not classified as publications were excluded; nested metadata structures (subjects, author identifiers) were flattened into tabular fields. The attributes retained for downstream analysis are the OpenAIRE identifier, publication date, and Field of Science (FOS) labels assigned by the SciNoBo classifier [5]. Full field specifications follow the OpenAIRE data model [3].

Citation links required a different approach: the OpenAIRE relationship dataset exceeds 7 billion records, so we processed the raw dumps via streaming rather than API calls, filtering for `Cites`/`IsCitedBy` entries [2] incident to at least one collected product. Bidirectional pairs were normalized to a single directed edge pointing from the citing paper to the cited one.

Citation graph construction

The raw citation network is a directed graph where edges point from the citing paper to the cited one. To focus on outgoing UniPi citations, we removed all edges originating from external nodes, retaining only UniPi-to-UniPi and UniPi-to-external edges.

A temporal filter—motivated by both analytical focus and computational feasibility—then reduced the UniPi node set to papers published from 2020 onward: *recent* nodes (36,809), plus older publications cited by at least one recent paper: *parent* nodes (26,053). A 2021 paper citing a foundational 2017 study keeps both in the graph; the 28 publications lacking a valid date are treated as parents when cited by a recent node. After removing isolates, the citation-only graph G_{Cit} contains 55,078 nodes and 217,174 undirected edges.

External nodes—all publications cited by any surviving UniPi paper—are not part of G_{Cit} but are retained for bibliographic coupling and hypergraph construction (Section 5). The full graph including these external nodes reaches 1,603,782 nodes and 2,694,706 edges.

Bibliographic coupling and edge weighting

Direct citation connects only papers that explicitly reference each other; two papers studying the same problem through the same sources remain unlinked if neither cites the other. Bibliographic coupling (BC) captures this missing proximity: two papers are coupled when they share external references.

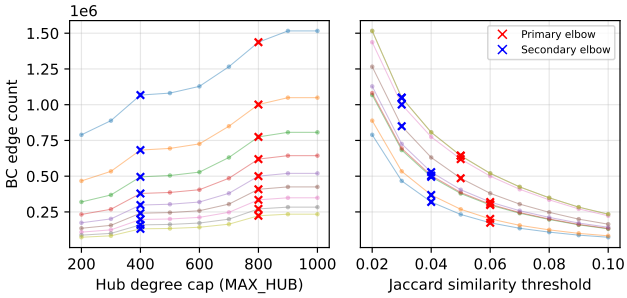


Figure 1: BC edge count under joint parameter sweep Left: varying MAX_HUB (one curve per Jaccard threshold). Right: varying Jaccard threshold (one curve per MAX_HUB). Crosses mark primary and secondary elbows per curve.

We quantify BC via the Jaccard similarity of external reference sets, and prefer it over co-citation because it is forward-looking—it requires no third-party citations and therefore applies to recent work.

Raw BC scores carry three kinds of noise. *Hub contamination*: a few foundational references cited across disciplines inflate BC pair counts combinatorially—we swept MAX_HUB from 200 to 1,000 in steps of 100 and set it to 800, the last stable step before edge count jumps sharply (Figure 1, left: primary elbows cluster at 800 across thresholds). *Weak coupling*: we swept the Jaccard threshold from 0.02 to 0.10 in steps of 0.01 and chose MIN_BC = 0.05, where primary elbows fall between 0.05 and 0.06 across hub caps (Figure 1, right). *Coincidental overlap*: a single shared reference is too little evidence, so MIN_SHARED = 2.

The final edge weight blends both signals:

$$w(u,v) = \alpha \cdot \mathbf{1}_{\text{direct}}(u,v) + (1-\alpha) \cdot \widetilde{\text{BC}}(u,v) \qquad (1)$$

where $\mathbf{1}_{\text{direct}}$ equals 1 if a direct citation exists, and $\widetilde{\text{BC}}$ is the Jaccard score normalized by its 99th percentile ($\text{BC}_{P99} = 0.2054$, preferred over the maximum to avoid outlier distortion).

The mixing parameter $\alpha = 0.35$ was selected through an InfoMap sensitivity analysis over $\alpha \in [0.10, 0.90]$. At this value the partition sits near a local NMI peak on both baselines (0.72 vs. directed, 0.84 vs. undirected citation-only partitions). Community count remains stable across the entire range (1,172–1,177), confirming that partition granularity depends on topology rather than weighting—α controls how much each signal contributes to edge weight, not how many communities emerge.

3 NETWORK CHARACTERIZATION

All metrics in this section use the undirected, unweighted projections of G_{Cit} and G_{BC} defined in Section 2; directed and weighted variants enter only when explicitly noted.

Table 1: Graph overview and connected-component analysis. Neither graph contains isolated nodes by construction.

	Full graph			LCC		
	N	M	CC	N	M	Coverage
G_{Cit}	55 078	217 174	1 488	49 243	209 599	89.4%
G_{BC}	57 603	745 451	1 168	53 405	739 498	92.7%
Δ	+2 525	+528 277	−320	+4 162	+529 899	+3.3 pp

Of G_{BC}'s edges, 70.9% come from BC alone, 19.5% from direct citation, and 9.7% carry both signals—so BC-only edges outnumber citation-only edges 3.64×. The structural effect is visible in Table 1: component count drops by 320 and LCC coverage rises by 3.3 pp, as shared references stitch together fragments that citation links leave isolated.

Degree Distribution

Table 2: Degree statistics on the undirected LCCs.

	Mean	Std	Median	Max	P90	P99	k=1 (%)
G_{Cit}	8.51	10.78	6	621	18	41	10.4
G_{BC}	27.69	103.21	8	1 169	30	726	7.8

BC more than triples average degree, but the effect concentrates in the upper tail: P99 jumps from 41 to 726 and standard deviation grows tenfold, while the median shifts only from 6 to 8. A typical node gains a handful of connections; a well-connected one gains hundreds. The fraction of degree-one nodes drops modestly (10.4% → 7.8%)—BC pulls some peripheral papers into denser neighbourhoods but does not eliminate the long tail of low-degree nodes.

Both distributions are approximately linear on a log-log CCDF (Figure 2), consistent with scale-free-like behaviour. G_{BC}'s fitted exponent ($\hat{\gamma} = 2.13$) is shallower than G_{Cit}'s ($\hat{\gamma} = 3.54$), confirming that high-degree nodes are proportionally more abundant when BC edges are included.

Clustering, Density, and Path Structure

Table 3: Clustering, density, and path structure on the undirected LCCs. C=0: fraction of nodes with zero clustering. Standard deviation is 0.284 for both graphs. L and diameter computed exactly via NetworKit.

	Clustering				Path structure	
	Avg C	Median	C=0 (%)	Density	L	Diam.
G_{Cit}	0.342	0.321	21.4	1.7×10^{-4}	11.81	48
G_{BC}	0.425	0.400	14.7	5.2×10^{-4}	10.04	33

BC triples density (3.06×) but raises average clustering by only 1.24×. Most BC edges connect papers that share references without belonging to the same local cluster—they bridge across neighbourhoods rather than close triangles within them. Zero-clustering nodes fall from 21.4% to 14.7%: BC does pull peripheral nodes into triangles, but without saturating local cohesion.

The weighted-to-unweighted clustering ratio $C_w/C_{uw} = 0.1782/0.4250 = 0.419$ is consistent with the Granovetter weak-tie effect: closed triangles are systematically composed of weaker ties than open ones, while inter-community bridges—the weak ties driving information diffusion—remain open triplets.

Those bridges do shorten paths. BC reduces mean geodesic by 15.0% (11.81 → 10.04) and diameter from 48 to 33—unlike clustering, both the bulk and the extremes of the distance distribution respond. Both averages remain high relative to the ER baseline ($L_{ER} = 3.65$ for G_{BC}, 5.30 for G_{Cit}), because paths traverse multiple community boundaries.

Comparison with Random Graph Models

Each real LCC is benchmarked against Erdős–Rényi $G(N, M)$—matching edge count exactly—and Barabási–Albert with $m = \text{round}(M/N)$ edges per step. The rounding constraint causes BA to slightly overshoot for G_{BC} ($M_{BC}^{BA} = 747{,}474$, +1.1%) and undershoot for G_{Cit} ($M_{Cit}^{BA} = 196{,}956$, −6.0%), but degree distributions and clustering are evaluated relative to the actual edge count, so the mismatch does not distort the comparison.

Table 4: Real LCCs vs null models. Path lengths for null models estimated via 500-node BFS sample (approximate); real graphs computed exactly. $\sigma = (C_{\mathbf{real}}/C_{\mathbf{ER}}) \,/\, (L_{\mathbf{real}}/L_{\mathbf{ER}})$.

		Graph				Degree		Clustering		Path length		$k_{\max}$		
		N	M	LCC	$\langle k \rangle$	γ	σ_k	C_{emp}	C_{th}	d_{emp}	d_{th}	$k_{\max}^{\text{emp}}$	$k_{\max}^{\text{th}}$	σ
G_{BC}	Real	53 405	739 498	53 405	27.69	2.13	103.21	0.4250	—	10.04	—	1 169	—	290.97
	ER	53 405	739 498	53 405	27.69	—	5.24	0.0005	0.00052	3.65	3.28	56	—	—
	BA	53 405	747 474	53 405	27.99	—	36.37	0.0037	—	3.20	—	1 317	—	—
G_{Cit}	Real	49 243	209 599	49 243	8.51	3.54	10.78	0.3424	—	11.81	—	621	—	808.71
	ER	49 243	209 599	49 231	8.51	—	2.91	0.0002	0.00017	5.30	5.05	24	—	—
	BA	49 243	196 956	49 243	8.00	—	12.14	0.0017	—	4.39	—	883	—	—

Clustering exceeds ER by roughly 800× for G_{BC} and 1,800× for G_{Cit}, while mean path length is only 2.2–2.8× longer—yielding $\sigma = 290.97$ and 808.71, far above the $\sigma > 1$ threshold. At citation-graph density, ER falls just below the full-connectivity threshold ($\langle k \rangle = 8.51 < \ln N \approx 10.8$), leaving 12 nodes outside the LCC; the real graph is fully connected at the same density because community organisation distributes edges non-randomly.

BA captures the heavy-tailed degree distribution (Figure 2) but not its exact shape. Its fixed exponent $\hat{\gamma}_{BA} \approx 3$ sits between the two empirical values in opposite ways: it overestimates for G_{BC} ($3 > 2.13$, predicting a lighter tail than

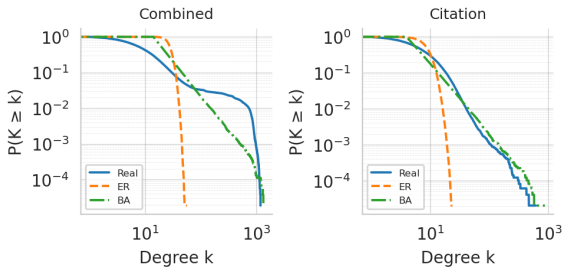


Figure 2: Complementary CDF of degree on a log-log scale for G_{BC} (left) and G_{Cit} (right), compared against ER and BA null models (dashed). The shallower slope of G_{BC} reflects its lower power-law exponent.

observed) and underestimates for G_{Cit} ($\beta < 3.54$, predicting a heavier tail). The $k_{\max}$ columns expose this directly: BA reaches 1 317 for G_{BC} against ER's 56, while the real maximum of 1 169 sits closer to BA—hubs are a genuine feature of the network, not an ER artefact. In the $\hat{\gamma} < 3$ regime, hubs are dense enough to act as routing shortcuts, which explains why BA achieves shorter paths than ER for G_{BC} ($d = 3.20$ vs 3.65). Neither null model accounts for the observed clustering; the residual is best explained by triadic closure driven by community membership, layered on top of preferential attachment.

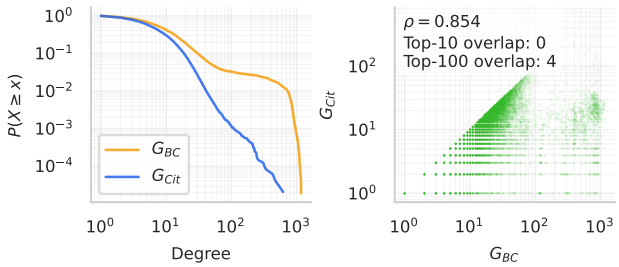
Centrality Analysis

Degree, betweenness (normalised), and closeness were computed exactly on both undirected LCCs via NetworKit.

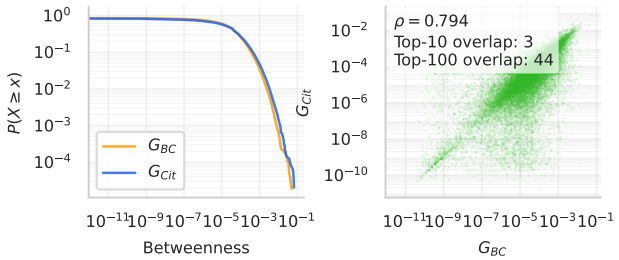
Table 5: Centrality statistics (exact, undirected LCCs). $> \mu+2\sigma$: nodes beyond two standard deviations above the mean.

	Mean	Std	Max	P99	$> \mu + 2\sigma$
Combined LCC					
Degree	27.69	103.21	1 169	726	1 391
Betweenness	1.69e-4	7.20e-4	0.0502	2.41e-3	970
Closeness	0.1017	0.0135	0.1435	0.1263	269
Citation LCC					
Degree	8.51	10.78	621	41	1 194
Betweenness	2.20e-4	1.02e-3	0.0666	3.38e-3	857
Closeness	0.0870	0.0131	0.1270	0.1107	180

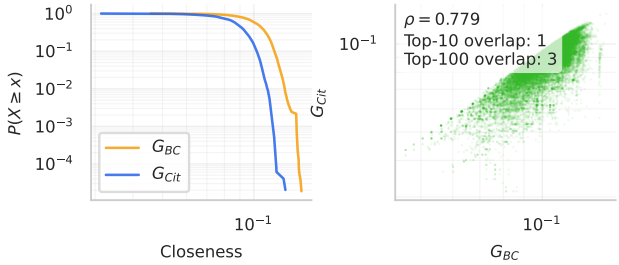
Denser connectivity shortens geodesics, which raises closeness ($\bar{c}$: 0.0870 → 0.1017) and simultaneously creates parallel routes that dilute betweenness ($\bar{b}$: 2.20 → 1.69×10^{-4}, maximum: 0.0666 → 0.0502). The $> \mu+2\sigma$ counts expose a related asymmetry: degree produces 1 391 outliers in G_{BC} and betweenness 970, while closeness yields only 269—proximity is far more uniformly distributed than either connectivity or brokerage. Well-connected nodes and well-positioned nodes are, to a large extent, distinct populations.



(a) Degree centrality.



(b) Betweenness centrality.



(c) Closeness centrality.

Figure 3: Centrality analysis on G_{BC} and G_{Cit}: log-log CCDF (left) and cross-graph scatter with Spearman ρ, top-10 and top-100 overlap (right), for degree, betweenness, and closeness.

Figure 3 captures two concurrent effects. Global rank order is well preserved ($\rho = 0.78$–0.85): a node that ranks high in G_{Cit} tends to remain high in G_{BC}. But the elite is almost entirely reshuffled. At top-10, degree overlap is zero and closeness just 1/10; even at top-100, both remain negligible (4 and 3 respectively). Betweenness is the exception: 3/10 at top-10 and 44/100 at top-100, suggesting that key brokers retain their structural role across both graphs. BC rewards papers with large, overlapping reference lists, not papers that attract citations—so collaboration gaps look different depending on which graph is analysed.

4 COMMUNITY DETECTION

BC reshapes global topology—density triples, path lengths shrink, and centrality rankings reshuffle almost entirely at the top (Section 3). But global metrics say nothing about *mesoscopic* organisation: do the communities themselves change, and if so, does the change carry disciplinary meaning? We address both questions by running three algorithms on G_{Cit} and G_{BC}, then validating the resulting partitions against FOS labels at two granularities (L2 macro-domains, L4 sub-disciplines).

Ensemble Stabilisation

All three algorithms are stochastic: different seeds yield different partitions. We stabilise each via an *ensemble-plus-medoid* protocol—run the algorithm N times, compute pairwise NMI across all runs, and select the partition with the highest mean NMI to all others (the medoid).

Leiden. [9] with `RBConfigurationVertexPartition` introduces an explicit resolution parameter γ, avoiding the resolution limit of standard modularity maximisation [4]. We sweep γ over 30 log-spaced values in [0.05, 10], running a 100-run ensemble at each point and recording the number of non-trivial communities ($\geq$ 20 nodes) in the medoid. Both graphs exhibit a plateau where community count is stable: $\gamma \in [0.05, 0.578]$ for G_{Cit}, [0.05, 0.472] for G_{BC}. We fix $\gamma_{\text{common}} = 0.154$ from the plateau intersection and apply it to both graphs—a trade-off that sacrifices per-graph optimality for comparability.

InfoMap. [8] minimises map-equation codelength for a random walker. No resolution parameter is needed—the map equation does not suffer from the Fortunato–Barthélemy limit. We run a 100-run ensemble and select the medoid. G_{Cit} enters as directed and unweighted; G_{BC} as undirected and weighted.

ANGEL. [7] is the only overlap-aware method, but its output is sensitive to two parameters: `threshold` (minimum similarity for merging ego-communities) and `min_community_size`. Stabilisation proceeds in two stages.

Parameter calibration. We sweep an 11×4 grid (threshold $\in$ [0.3, 0.8], min_size $\in$ [4, 7]), running each configuration on both graphs. A robust utility function combines distance-to-ideal and maximin criteria across coverage, overlap rate, and community count. The selected configuration is `threshold` = 0.45, `min_community_size` = 4.

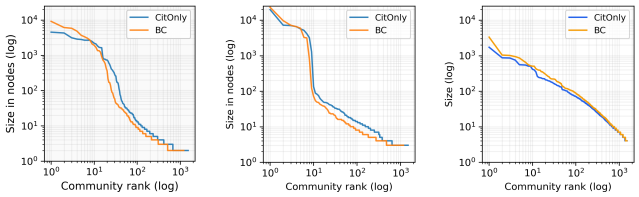
Ensemble. The selected configuration is run 30 times (reduced from 100 due to computational cost: ~640 s per BC run). The medoid is selected via NMI on crisp projections, where each multi-member node is assigned to the community containing the most of its neighbours.

Stability. Table 6 confirms that all methods are highly stable ($\mu_{\text{NMI}} \geq 0.91$). InfoMap on G_{BC} shows the widest spread ($\sigma = 0.037$)—the denser graph creates more flow-routing ambiguity for the random walker.

Table 6: Ensemble stability: pairwise NMI across all runs measures inter-run agreement. Higher μ_{NMI} means the algorithm produces near-identical partitions from different seeds.

Algorithm	Graph	Runs	μ_{NMI}	σ	Medoid
Leiden	G_{Cit}	100	0.941	0.007	0.952
Leiden	G_{BC}	100	0.934	0.011	0.946
InfoMap	G_{Cit}	100	0.941	0.009	0.954
InfoMap	G_{BC}	100	0.910	0.037	0.934
ANGEL	G_{Cit}	30	0.967	0.002	0.969
ANGEL	G_{BC}	30	0.964	0.002	0.966

Algorithm	Graph	# Comm.	Med.	Mean	Largest	μ_{NMI}
Leiden	G_{Cit}	1 533	2	35.9	4 520	0.941
Leiden	G_{BC}	1 210	2	47.6	9 145	0.934
InfoMap	G_{Cit}	1 496	3	45.7	19 750	0.941
InfoMap	G_{BC}	1 174	3	55.4	22 940	0.910
ANGEL	G_{Cit}	1 527	9	25.4	1 724	0.967
ANGEL	G_{BC}	1 514	10	31.3	3 321	0.964



(a) Leiden **(b) InfoMap** **(c) ANGEL**

Figure 4: Community detection summary (medoid partitions) and community size distributions. Leiden: $\gamma = 0.154$, 100 runs. InfoMap: 100 runs. ANGEL: threshold = 0.45, min_size = 4, 30 runs.

Community Structure and Quality

FOS L4 labels cover 44 317 of 57 603 nodes (76.9%); the remaining 23.1% introduces uncertainty in every disciplinary classification that follows. The corpus is biomedical-dominant: medical and health sciences account for 49% of labelled nodes, followed by natural sciences (36%) and engineering (21%; percentages exceed 100% because nodes carry multiple labels).

Leiden and InfoMap agree: BC reduces community count by ~21% (Figure 4). ANGEL barely moves (−0.9%) because it resolves ambiguity through overlap rather than merging—its largest community nearly doubles (1 724 → 3 321), but the total count stays flat. Median size stays at 2–3 for the crisp methods on both graphs, reflecting the heavy tail typical of citation networks: most communities are tiny and unaffected by BC. The consolidation happens in the upper tail—Leiden

Table 7: Merge consensus. Median overlap fraction per algorithm.

Category	Merges	Med. InfoMap	Med. ANGEL
INTRA	3	0.99	0.30
CROSS	15	0.98	0.14
Total	18	—	—

needs 9 communities to cover 50% of nodes on G_{Cit} but only 5 on G_{BC}.

A fate analysis on communities with $\geq$ 20 nodes tracks what happens to each G_{Cit} community on G_{BC}. Leiden identifies 18 merge events. InfoMap merges more aggressively, but concentrates its merges into only 5 events—one of which absorbs 10 components into a single 19 664-node super-community. ANGEL stays largely stable (76%), absorbing change through overlap. Modularity remains above 0.6 on G_{BC} for both crisp methods ($Q = 0.681$ for Leiden, 0.597 for InfoMap), confirming that community structure retains measurable quality after BC.

The InfoMap super-community deserves scrutiny. At 19 664 nodes it covers roughly 36% of the network—a partition that assigns a third of all papers to a single group has limited discriminative value. Its internal FOS distribution is near-uniform (median $\hat{H} = 0.81$), meaning it groups papers across disciplines without thematic selectivity. BC consolidation pushes it further to 22 940 nodes, but the community was already too coarse on G_{Cit} to carry disciplinary meaning.

Semantic purity is measured as normalised Shannon entropy $\hat{H} = H/\log_2 k$ over FOS labels per community ($\geq$ 10 nodes); $\hat{H} \to 0$ is pure, $\hat{H} \to 1$ is uniform. Leiden's L4 entropy rises slightly on G_{BC} ($\bar{H} : 0.612 \to 0.631$)—consistent with cross-disciplinary merges broadening composition. InfoMap moves in the opposite direction ($\bar{H} : 0.683 \to 0.627$): its G_{Cit} partition contains a few giant communities with near-uniform FOS distributions, and BC-induced consolidation replaces them with tighter clusters. Neither shift is large enough to indicate degradation.

Cross-Algorithm Synthesis

For each of the 18 Leiden merge events (components $\geq$ 20 nodes), we check whether InfoMap and ANGEL independently group the same nodes together. An algorithm "confirms" a merge if $\geq$ 50% of the merged node set falls within one of its communities (Table 7).

InfoMap confirms almost every Leiden merge (median overlap $\geq$ 0.98). ANGEL does not—only 1 of 18 merges reaches 3/3 consensus. The gap is structural: ANGEL distributes boundary nodes across multiple communities rather than collapsing them, so its low overlap fraction (median 0.14 for CROSS) reflects fragmentation, not disagreement. Fifteen of 18 merges (83%) are cross-disciplinary at FOS L4.

To test whether merge zones are more disciplinarily concentrated than their surroundings, we compute Gini impurity $G_I = 1 - \sum_k p_k^2$ and compare it to a multi-hop neighbourhood null (1 000 permutations).

At hop = 1, the neighbourhood pool is too small to test any merge. At hop = 2, four of seven testable merges reach significance ($p < 0.05$), all CROSS. At hop = 3, 13 of 16 testable merges are significant—12 CROSS and 1 INTRA. BC creates structures that become distinguishable from their surroundings only at mesoscopic scale; locally, they blend into the same disciplinary landscape.

A complementary FOS gap analysis counts how often subdiscipline pairs co-occur within the same community in G_{Cit} vs G_{BC}. 206 pairs show a positive gap; 55 are entirely new. But a permutation test (1 000 shuffles preserving community sizes) finds none significant—all p-values equal 1.000. This is the key negative result of the community analysis: when communities are large (3,000–9,000 nodes), any size-matched random partition captures the same co-occurrence patterns. The FOS gaps exist structurally but carry no more disciplinary signal than chance given community size.

Caveats. The 83% CROSS ratio is conditioned on $\gamma = 0.154$, which sits at the lower end of the plateau and produces relatively large communities. Larger communities are more likely to span multiple FOS L4 labels *before* any merge, biasing classification toward CROSS. At higher γ, the same consolidation would register as more INTRA. The direction of BC's effect—fewer communities, larger merge zones—is robust across the plateau; the INTRA/CROSS proportion is resolution-dependent, not an absolute structural property. Three further limitations apply: 23.1% of nodes lack FOS labels; the shared γ_{common} sacrifices per-graph optimality; and ANGEL's 30-run ensemble provides less stabilisation than the other methods, though its tight NMI spread ($\sigma = 0.002$) suggests this is not limiting.

5 HYPERGRAPH ANALYSIS

Pairwise edges—whether from direct citation or bibliographic coupling—encode dyadic proximity. But when three papers all cite the same external reference, the three pairwise links that result are not independent: they share a common cause. A hypergraph captures this distinction natively. We define $H = (V, \mathcal{E})$ where V is the set of UniPi papers and each hyperedge $e \in \mathcal{E}$ groups all UniPi papers that cite the same external reference. The construction deliberately excludes direct citations between UniPi papers so that the hypergraph isolates the bibliographic-coupling signal without double-counting edges already in G_{BC}.

Table 8: S-line graph statistics. Citation LCC giant ratio: 0.894 **(dashed baseline).**

s	Edges	Components	Giant ratio
1	46,442,020	5,966	0.988
2	6,684,734	136,873	0.715
3	1,776,668	337,476	0.309
4	650,389	424,437	0.133
5	290,806	463,630	0.060

Construction

Starting from the full OpenAIRE citation graph (4.86M nodes, 8.27M edges), we retain post-2020 UniPi papers, their parent papers (older UniPi works cited by recent ones), and the external references they connect to—64,156 surviving UniPi nodes, 3.01M external nodes, 5.03M edges after removing external→external links.

For each external reference, we collect all UniPi papers connected to it (in either citation direction). References reaching fewer than 2 UniPi papers are discarded: they cannot form a hyperedge. This yields 762,800 raw hyperedges.

Deduplication. Multiple external references sometimes connect the exact same set of UniPi papers—semantically redundant hyperedges. We collapse identical node-sets into weighted "families," reducing the count to 512,004 unique hyperedges (duplicate rate: 32.9%). The most duplicated family has weight 1,535.

Hub filter. Foundational references cited across many disciplines generate spurious connections, just as in pairwise BC. We remove hyperedges with more than 450 UniPi nodes, eliminating 14 families and leaving 511,990 hyperedges over 61,369 nodes. Median hyperedge size is 3 (mean 3.9, max 423); median node degree is 19 (mean 32.2, max 6,061). The largest connected component covers 99.4% of nodes across 141 components—only 2 have ≥ 10 members.

S-Line Sensitivity

The s-line graph $L_s(H)$ connects two hyperedges whenever they share $\geq s$ nodes. At $s = 1$, any shared node suffices; as s grows, only strong higher-order overlaps survive. We measure sensitivity via the *giant ratio*—the fraction of hyperedges in the largest connected component of $L_s(H)$—and compare it to the citation baseline (LCC of G_{Cit}: 89.4%).

Table 8 shows the result. At $s = 1$, the giant ratio is 98.8%—nearly everything connects. It drops to 71.5% at $s = 2$ and to 30.9% at $s = 3$. The crossover with the citation baseline falls near $s = 2$: sharing two or more external references is more selective than a direct citation link. Beyond $s = 3$ the hypergraph fragments rapidly; at $s = 10$ only 0.1% of hyperedges remain in the giant component.

Table 9: Top 10 gap candidates by debiased score. "Support" is the number of shared hyperedges; FOS is the L2 domain. N/A indicates unlabelled nodes (23.1% of the corpus).

Rank	Support	Score	FOS pair
1	417	16.73	Clinical med. / Clinical med.
2	197	11.94	Clinical med. / N/A
3	163	11.30	Physical sci. / Physical sci.
4	259	11.15	Clinical med. / N/A
5	255	10.93	Clinical med. / Clinical med.
6	211	10.35	Physical sci. / Physical sci.
7	140	10.10	Physical sci. / Physical sci.
8	144	9.94	Clinical med. / N/A
9	171	9.78	Clinical med. / N/A
10	482	9.19	Physical sci. / Physical sci.

Two implementation constraints apply. Hyperedges with > 80 nodes are excluded from overlap computation (too many pairwise comparisons), and nodes with hyperdegree > 300 are filtered before building the overlap matrix. These thresholds remove 308 nodes and 235 hyperedges, leaving 511,755 working hyperedges over 61,342 nodes.

Gap Candidates

A *gap candidate* is a pair of UniPi papers that share ≥ 1 hyperedge but have no direct citation between them. From the overlap matrix, 2,000,243 such pairs exist. Their support distribution is heavy-tailed: the Q90 threshold is 4 shared hyperedges, and the top pair shares 1,927.

Raw support rankings are dominated by high-degree nodes. We debias with the formula

$$\text{score}_{\text{debias}}(u, v) = \frac{|\mathcal{E}_u \cap \mathcal{E}_v|}{\sqrt{(d_u + 1)(d_v + 1)}} \tag{2}$$

where d_u, d_v are citation-graph degrees (missing degrees imputed at the median). The correction is severe: Jaccard overlap between the raw and debiased top-30 is 0.111. Without debiasing, rankings reflect hub proximity rather than intellectual affinity.

The top debiased pair (score = 16.73) shares 417 hyperedges between two clinical medicine papers with mean citation degree 24. Table 9 lists the top 10. Clinical medicine and physical sciences dominate, reflecting the corpus composition noted in Section 4.

Null Model Validation

Do gap candidates reflect genuine intellectual proximity, or could the same co-occurrence pattern arise from degree structure alone? We test with a degree-preserving null model: `double_edge_swap` on G_{Cit} (200 simulations), measuring citation hit rate for the top 2,000 pairs by raw hyperedge support.

The observed hit rate is 69.1%—over two-thirds of these pairs, despite lacking a direct citation, have one in the randomised baseline. The null mean is 1.3% (±0.24%), yielding $z = 284.5$ ($p < 0.01$). Hyperedge co-occurrence predicts citation probability far beyond what degree distribution can explain.

This is the strongest quantitative result of the hypergraph analysis. It validates the construction: the signal in shared external references is not an artefact of hub nodes or degree heterogeneity.

Convergence with Pairwise Results

We connect hypergraph gaps back to the Leiden partitions from Section 4. For each of the top 100 debiased gap candidates, we check two conditions: (1) whether both nodes changed community assignment between G_{Cit} and G_{BC} (merge zone), and (2) whether both nodes belong to the same Leiden community on G_{BC}.

92% of top-100 gap candidates share the same Leiden community on G_{BC}, against a random baseline of 0.3%. The merge-zone statistic is uninformative: 100% of gap candidates fall in merge zones, but so do 100% of random pairs (1.0× enrichment). At $\gamma = 0.154$, community reassignment is so pervasive that merge-zone membership is not discriminating.

The same-community convergence, however, is strong. Two independent representations—pairwise BC communities and hyperedge co-occurrence—place the same paper pairs together. Neither method was designed to agree with the other; the convergence is a structural property of the underlying citation data.

Semantic Profile of Top Gaps

Of the top 50 debiased gap candidates, 36 (72%) connect papers in the same FOS L2 domain, and 38 (76%) share at least one L2 domain. Twelve pairs (24%) span different domains entirely—these are the most interesting for cross-disciplinary collaboration discovery.

The dominance of same-domain pairs is consistent with the community-detection finding that BC primarily consolidates intra-disciplinary fragments (Section 4). Cross-disciplinary gaps exist but are the minority, not the majority. FOS coverage limitations apply here as well: 4 of the 12 cross-disciplinary pairs involve at least one N/A node, so the true cross-disciplinary count may be lower.

Limitations

Several constraints bound the analysis. Hyperedges with > 80 nodes were excluded from overlap computation, and nodes with hyperdegree > 300 from s-line construction—both to keep computation tractable, but both remove exactly the highest-connected structures where cross-disciplinary overlap is most likely. Gap support counts *node pairs*, not unique intellectual connections: a single prolific author appearing in both papers inflates the count.

The null model validates citation prediction but does not test whether gap candidates represent *actionable* collaboration opportunities. A shared reference list implies thematic overlap; it does not imply that the authors would benefit from working together. That inference requires domain expertise beyond what network analysis can provide.

6 CROSS-METHOD SYNTHESIS

Community detection (Section 4) and hypergraph analysis (Section 5) attack the same question from independent angles. Neither method was designed to agree with the other. Where they converge, the evidence is structural; where they diverge, the gap is informative.

Where the Methods Agree

Three findings hold across both representations. BC and hyperedge co-occurrence both act primarily within disciplinary boundaries, stitching together fragments that direct citation leaves disconnected (Sections 4, 5). Cross-disciplinary bridges exist but are not the dominant mode of restructuring. And the top hypergraph gap candidates cluster within the same Leiden communities on G_{BC}—though this convergence is partly expected, since both representations derive from shared external references (Section 5).

Where the Methods Diverge

Scale of evidence. Community detection operates at the mesoscopic level: merge events involve tens to thousands of nodes, and significance emerges at multi-hop scale. The hypergraph operates at the microscopic level: gap candidates are individual paper pairs, validated against a degree-preserving null. The two methods answer different questions—"which communities restructure?" vs. "which pairs are suspiciously close without citing each other?"—and the answers are complementary rather than redundant.

Sensitivity to degree. Debiasing radically reshuffles hypergraph rankings (Jaccard = 0.111 between raw and debiased top-30), while community-detection results are less sensitive to individual node degrees. Any downstream use of gap candidates must apply degree correction; raw support counts are dominated by hub proximity.

Overlap vs. crisp boundaries. ANGEL detects overlap nodes that Leiden and InfoMap must assign to a single community. The hypergraph does not partition nodes at all—it ranks pairs. When a paper sits at the intersection of two communities, ANGEL flags it as a bridge; the hypergraph flags its strongest uncited neighbours as gap candidates. Both capture boundary phenomena, but through different lenses.

What the Evidence Supports—and What It Does Not

The combined evidence supports three claims. First, bibliographic coupling captures thematic proximity that direct citation misses—density grows, path lengths shrink, and communities consolidate. Second, this proximity is predominantly intra-disciplinary: BC connects researchers working in the same broad area who happen not to cite each other, not researchers in unrelated fields. Third, the hypergraph independently confirms this at the pair level, with a citation-prediction signal that rules out degree-driven artefacts.

The evidence does *not* support the claim that any specific gap candidate represents a "missed collaboration." Thematic proximity is necessary but not sufficient: two papers may share references because they address the same well-studied problem from independent perspectives, or because a foundational reference is universally cited within a subfield. Converting gap candidates into actionable recommendations requires author-level disambiguation, temporal ordering, and domain-expert validation—none of which falls within the scope of this analysis.

7 CONCLUSIONS

We constructed two graph representations and one hypergraph from University of Pisa publications in the OpenAIRE Research Graph, then applied three community-detection algorithms and a higher-order gap analysis. Bibliographic coupling consolidates community structure—predominantly within disciplines—and the hypergraph independently confirms this proximity at the pair level. The strongest quantitative signal is the citation-prediction null model ($z = 284.5$), which rules out degree-driven artefacts; the weakest is the FOS co-occurrence permutation test ($p = 1.000$), where community size alone explains the observed disciplinary patterns.

Known Limitations

Method-specific caveats—resolution dependence of the INTRA/CROSS ratio, ANGEL's reduced ensemble, shared γ_{common}—are discussed in Sections 4 and 5. The limitations below are structural.

Data completeness and labelling. The OpenAIRE Research Graph, while extensive (345M+ records), reflects only what has been indexed and deduplicated. 23.1% of UniPi nodes lack FOS labels entirely, and this fraction is not uniformly distributed across fields—newer or more applied areas may be systematically under-labelled. Every disciplinary classification in this report carries that uncertainty. Author metadata is similarly incomplete: affiliations, ORCID links, and disambiguated author identities are missing for a non-trivial fraction of records, which is why we operate at the paper level rather than the author level.

Temporal filtering bias. Retaining only post-2020 publications (plus cited parents) was both a methodological choice and a computational constraint. The filter biases the corpus toward fields with high recent output—clinical medicine and physical sciences dominate, while slower-publishing humanities are under-represented. Older but potentially relevant citation patterns are excluded entirely. A 2018 paper that would have bridged two communities is invisibleto this analysis unless a post-2020 paper cites it.

Bibliographic coupling design choices. Several BC parameters were fixed via sensitivity sweeps but not revisited after downstream results were available. The hub cap (MAX_HUB = 800) and Jaccard threshold (MIN_BC = 0.05) were chosen at elbow points in edge-count curves, but different downstream tasks—community detection vs. gap candidate ranking—might benefit from different thresholds. More fundamentally, we never constructed a *pure* BC graph (bibliographic coupling edges only, no direct citations) to compare against G_{Cit}. The current G_{BC} blends both signals via the α-weighted formula, which means community differences between G_{Cit} and G_{BC} confound two effects: the addition of thematic-proximity edges and the re-weighting of existing citation edges. Disentangling the two would require a three-way comparison (G_{Cit} vs. $G_{\text{BC-only}}$ vs. G_{BC}) that was not performed.

Hypergraph filtering and scale. Excluding hyperedges with > 80 nodes and nodes with hyperdegree > 300 from the overlap matrix was necessary for tractability. These thresholds remove the highest-connected structures—precisely the ones where cross-disciplinary overlap is most likely. The hub filter (MAX_HUB = 450) uses a different threshold than the pairwise BC hub cap (800), introducing an inconsistency between the two representations. No systematic study of how hypergraph gap rankings change under different hub thresholds was conducted.

Single-level analysis on a multi-level graph. The full citation graph contains 1.6M nodes and 2.7M edges spanning UniPi papers, their parents, and the external references they cite. Yet all community detection and centrality analysis operates on the UniPi-only projection—a single-level slice that discards the intermediate structure connecting UniPi papers through external nodes. Bibliographic coupling recovers part of this structure by encoding shared references as pairwise edge weights, but it collapses multi-hop paths into a single number. The hypergraph does better—each hyperedge preserves which external reference creates the connection—but still treats all external nodes as interchangeable anchors rather than as nodes with their own neighbourhood structure.

No ground-truth validation. Gap candidates identify thematic proximity, not actionable collaboration. Without retrospective co-authorship data or domain-expert review, the pipeline's practical value remains untested.

Future Directions

Retrospective validation. The most direct test of the pipeline's predictive value is retrospective: flag gap candidates in 2020 data, then check for co-authorship by 2025. Such a design would yield ground-truth precision and recall estimates. Positive cases validate the method; negative cases may reveal systematic biases—for instance, whether the pipeline overweights large labs that share references due to shared supervisors rather than shared intellectual interests.

Pure bibliographic coupling graph. Comparing Leiden partitions across G_{Cit}, $G_{\text{BC-only}}$ (coupling edges only, no direct citations), and the blended G_{BC} would isolate the contribution of thematic-proximity edges from the re-weighting effect. If community merges persist in $G_{\text{BC-only}}$, the thematic signal is genuine; if they vanish, the merges are driven by the blending formula rather than by shared references.

Multi-hop and multi-level analysis. Exploring two- and three-hop paths through external nodes—asking not just "do A and B share a reference?" but "do A and B share references that are themselves co-cited in the broader literature?"—would capture deeper thematic affinity than pairwise BC or single-layer hyperedges can express. Multi-level community detection methods operating natively on bipartite or k-partite graphs would preserve the external-node layer as a first-class structural element rather than projecting it away.

Author-level resolution. Lifting to author-level networks would enable structurally grounded recommendations: "researcher A and researcher B share 15 reference clusters but have never co-authored." The principal technical challenge is scalability: author pairs grow quadratically with papers per author, and prolific researchers would dominate gap rankings even after degree correction.

Temporal dynamics. Sliding-window community detection—tracking merge and split events over yearly slices—would reveal whether BC-induced bridges *precede* actual co-authorship, turning the pipeline into a form of temporal link prediction. The temporal dimension would also allow filtering gap candidates by activity overlap, removing pairs where one author stopped publishing before the other became active.

Finer disciplinary taxonomies. Replacing FOS with the Italian *Settori Scientifico-Disciplinari* (SSD), journal-level subject categories, or embedding-based topic models from abstracts would sharpen both community labelling and gap profiling. The 23.1% unlabelled fraction could be partially recovered through abstract-based classification.

Weighted and directed hypergraph extensions. Weighting hyperedges by inverse document frequency would down-weight universally cited works and amplify niche-specific shared references. A directed variant, encoding temporal precedence or citation direction, could distinguish "papers that build on the same foundation" from "papers that are built upon by the same downstream work"—a distinction the current undirected construction collapses.

Multi-institution comparison. Extending the pipeline to inter-institutional analysis—comparing UniPi with other Italian universities—would test whether BC-driven consolidation generalises beyond this corpus. Cross-institution gaps suggest not internal collaboration but potential inter-institutional partnerships or co-supervision opportunities.

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